

Triangulations

$X \rightarrow$ triangulable, compact metric space

↳ By triangulation we mean a finite simplicial complex K together with a homeomorphism $\phi: \|K\| \rightarrow X$

↳ geometric realization of K by assigning coordinates to vertices & gluing faces together accordingly

for simplex $\sigma \in K$,

$$\text{diam}(\sigma) := \max_{x, y \in \sigma} d(x, y)$$

$$\text{mesh}(K) = \max_{\sigma \in K} \text{diam}(\sigma)$$

$$\text{size}(\text{triangulation of } K) = |K| \text{ i.e. no. of simplices}$$

If $\dim X = n$, then K consists of simplices of dim b/w 0 & n .

for $0 \leq l \leq n$, the l -skeleton $K^{(l)}$ is the subset of simplices of dim $\leq l$ for $x \in \mathbb{R}^+$

$$N(x) := \min_{\text{mesh}(K) \leq x} |K| \text{ and } N_l(x) := \min_{\text{mesh}(K) \leq x} (|K^{(l)}| - |K^{(l-1)}|)$$

$$\text{Clearly, } \sum_{k=0}^n N_k(x) \leq N_n$$

Example: 1) $X = S^n$

for $n=1$, its length is 2π and we can decompose it into $m = \lceil \frac{2\pi}{x} \rceil$ arcs of length $\leq x$ each. Assuming $x < \pi$, we can triangulate with wedges and m vertices. Not possible to do better and so

$$N_0(x) = m \text{ \& } N_1(x) = 2m - m = m \text{ \& } N(x) = 2m \text{ (}\because |K| = m + m\text{)}$$

for $n \geq 2$, $\exists c, C > 0$ that depend on n & n -dim Volume of S^n s.t

means that $\frac{c}{x^n} \leq N(x) \leq \frac{C}{x^n}$
no. of simplices in triangulation of mesh $\leq x$ grows like $\frac{1}{x^n}$.

Reason: $\text{mesh} \leq x \Rightarrow \text{diam}(\sigma) \leq x$ for each σ in triangulation of K

In n dim, $\text{diam}(\sigma) \leq x \Rightarrow \text{Vol} \leq x^n$

$$\therefore \frac{\text{Vol}(S^n)}{\text{Vol}(\sigma)} \approx \frac{1}{x^n}$$

2) If X is a cpt Riemannian n -manifold, then N exhibits similar behaviour for small ϵ but with significantly different behaviour for large values of ϵ .

Take $X = 2$ dim Torus

$$= \{x \in \mathbb{R}^3 : d(x, C_\epsilon^{(3)}) = \epsilon\}$$

major radius $\uparrow$ $\hookrightarrow$ circle of radius $\frac{\epsilon}{2}$ in $\mathbb{R}^3$

$$\text{Area}(X) = 4\pi^2 \cdot \frac{1}{\epsilon} \cdot \epsilon = 4\pi^2$$

Then for $\epsilon \leq \epsilon$, $N(\epsilon) \propto \frac{1}{\epsilon^2}$ & for $\epsilon > \epsilon$ each Δ covers $\sim \epsilon^2$ area & so $N(\epsilon) \propto \frac{\text{Area}}{\epsilon^2} \sim \frac{1}{\epsilon^2}$

$$\epsilon > \epsilon, N(\epsilon) \propto \frac{1}{\epsilon \epsilon}$$

$\hookrightarrow$ triangulate with rectangular strip of length $\sim \frac{\epsilon}{\epsilon}$ & width $\sim \epsilon$. To tile this with r s.t. $\text{diam}(r) \leq \epsilon$, need $\sim \frac{1}{\epsilon}$ tiles along long direction & $\sim \frac{1}{\epsilon}$ along short

Snapping

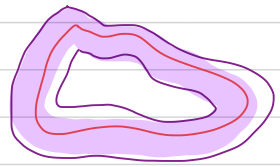
let $K \rightarrow$ triangulation of X with $\text{mesh} \leq \epsilon$

for $z \in X$, consider thickened version $z^\epsilon = \{x \in X : \exists y \in z \text{ with } d(x, y) \leq \epsilon\}$

Snapping Lemma: let $K \rightarrow$ triang of a cpt metric space X with $\text{mesh}(K) = \epsilon$. Then for each cycle z of dim l in X ,

$\exists$ cycle $\bar{z}$ in the l -skeleton of K that is homologous to z inside z^ϵ . Two k -cycles $c, c' \in C_k(K)$ are homologous if

bound same hole $c - c' = \partial_{k+1}(d)$ for some $d \in C_{k+1}$



Pf:

let $K' \rightarrow$ minimal subcomplex of K s.t. $\|K'\|$ contains z . (If top space X is trian,

$\therefore$ the map that sends each simplicial cycle of K' to its associated singular cycle induces an isomorphism at the $H_2^{\text{sing}}(K') \cong H_2^{\text{sing}}(K)$ i.e.

simplicial & singular homology

homology level, $\therefore z$ is homologous to a simplicial cycle $\bar{z}$ (groups are isomorphic)

within K' . $\therefore$ All $\sigma \in K'$ have $\text{diam} \leq \epsilon$ and K' consists only of simplices whose geometric images lie inside a union of balls of radius ϵ ,

$$\|K'\| \subset z^\epsilon$$

QED ($\because \bar{z} \sim z$ inside $\|K'\|$, chain w.r.t. $\partial w = z - \bar{z} \in \|K'\| \subset z^\epsilon$)

Lipschitz fns: $\exists c: f(x) - f(y) \leq c d(x, y)$

Lipschitz constant of f ($\text{Lip } f$) = inf over all such c

Separation lemma: The distance b/w the level sets defined by values $a \leq b$ is at least the diff in values divided by the Lipschitz constant, i.e.,

$$d(x, y) \geq \frac{b-a}{\text{Lip}(f)} \text{ whenever } f(x)=a \text{ \& } f(y)=b$$

(Implies that the sublevel set X_b contains the thickened version X_a^ϵ consisting of all pts at dist $\leq \epsilon = \frac{b-a}{\text{Lip } f}$

$$\text{Amp}(f) := \max_{x \in X} f(x) - \min_{y \in X} f(y)$$

(Clearly, $\text{Amp}(f) \leq \text{diam}(X) \cdot \text{Lip}(f)$

Persistent Homology

$X \rightarrow$ cpt metric space & $f: X \rightarrow \mathbb{R}$ is Lipschitz

$$X_a := f^{-1}((-\infty, a]) \rightarrow \text{sublevel}$$

$$\text{so } a \leq b \Rightarrow X_a \subseteq X_b$$

$$\therefore \exists \text{ homomorphism } f_x^{a,b}: H_k(X_a) \rightarrow H_k(X_b) \rightarrow \text{as defined in course}$$

We assume that homology groups are defined for modulo 2 arithmetic

The nested family of sublevel sets defines a sequence of homology groups connected by the described homomorphisms.

f is tame if this seq is finite & consists of homology grps with $< \infty$ rank.
(finitely many critical values, fin dim homology, persistent diagrams are finite

Def: We say a class $\alpha \in H_k(X_a)$ is born at a if $\alpha \notin \text{Im } f_x^{a-s, a}$ for any $s > 0$.

We say α born at a dies entering b if $f_x^{a, b-s}(x) \notin \text{Im } f_x^{a-s, b-s}$ for any $s > 0$

but $f_x^{a, b}(x) \in \text{Im } f_x^{a-s, b}$.

Notation: α is born at a & dies at $b \rightarrow b(x) = a, d(x) = b$.

$$x = \sigma^1$$

$$f(x, y) = y$$

$$X_a = \{ (x, y) \in \mathbb{R}^2 : y \leq a \}$$

$$\text{for } a < -1, X_a = \emptyset$$

as $a \uparrow$, X_a grows from bottom to top of circle

at $a = 0$, half $\mathbb{R}^2$ & at $a = 1$, full circle

On H_1 , loop is not formed until we reach the entire circle & so a 1-dim class d is born near $a = 1$ & never dies as filtration ends at the whole space

In case x does not die in the seq which ends with $H_k(X)$, we set

$$d(x) = \max_{x \in X} f(x)$$

Def: The persistence of the class is $\text{pers}(d) = d(b) - b(d)$

Persistence diagrams

Represent births & deaths of ℓ^{th} homology class by set of pts in $\mathbb{R}^2$, the ℓ^{th} persistence diagram, denoted by $\text{Dgm}_\ell(f)$

for each ℓ -dim class d , $\text{Dgm}_\ell(f)$ consists of a pt $x = (b(x), d(x))$.

$$\text{pers}(x) = b(x) - d(x) \rightarrow \text{persistence of that pt}$$

Remark: for class d with $b(d) = a$, $d(d) = b$, the entire coset of homology classes is born at a & dies at b .

Specifically,

every class $d + \beta$ with β in $\text{Im } f_x^{a \rightarrow b, a}$ is in the coset.

The point (a, b) in $\text{Dgm}_\ell(f)$ represents all these classes. It is a single pt showing that the rank of ℓ^{th} homology group goes up by 1 at a and drops by 1 at b .

This is the case in which the rank changes by the appearance or disappearance of a single generator.

In the degenerate case, we may have > 1 generators at the same critical pt.

Say $a = b(d_1) = \dots = b(d_k)$ and all d_i independent. Then classes born at a are of the form $d + \beta$ with d is a non-zero combination of d_i & $\beta \in \text{Im } f_x^{a \rightarrow b, a}$. The rank of the ℓ^{th} homology group increases by k . The d_i 's may die at the same or at diff critical values. (f.e. $d(d_1) \leq d(d_2) \leq \dots \leq d(d_k)$) Then for each combination $d = \sum x_i d_i$, we have a largest index $k = k(d) \geq 1$ s.t. $x_k = 1$. Then $d + \beta$ dies at $d(d_k)$, the critical value at which its last constituent dies. Similarly we can have arbitrary no. k of independent classes die at the same value b . Thus the rank of H_ℓ drops by k & the total multiplicity of points with death coordinate b is k .

Persistent cycles

no. of ind ℓ -dim homology classes born at a & dead at b

Recall - pts in $D_{\text{pse}, \varepsilon}(f)$ have integer multiplicities

We consider a point with multiplicity k as k individual pts

Persistent Cycle Lemma: Let X be a triangulable, cft metric space & $f: X \rightarrow \mathbb{R}$ a tame Lipschitz fn. Then the no. of pts in the persistence diagrams of f whose persistence exceeds ε is at most $N(\frac{\varepsilon}{\text{Lip} f})$

Pf:

Let $\alpha = \frac{\varepsilon}{\text{Lip} f} \in K \rightarrow$ triang. of X with $\text{mesh}(K) \leq \alpha$ and no. of ℓ -simplices equal to $N_\ell(\alpha)$

let α be an ℓ -dim homology class s.t. $\text{pers}(\alpha) = d(\alpha) - b(\alpha) > \varepsilon$

By def of birth, $\exists$ a cycle $z(\alpha)$ in $X_{b(\alpha)}$ that gens α in $H_\ell(X_{b(\alpha)})$.

By Separation lemma,

$$X_{b(\alpha)+\varepsilon} > X_{b(\alpha)}^{\alpha}$$

$\rightarrow$ if $x \in X: d(x, X_{b(\alpha)}) < \frac{\varepsilon}{\text{Lip} f}$, then $\exists y \in X_{b(\alpha)}$ s.t. $\hookrightarrow f(y) \leq b(\alpha)$

$$d(x, y) < \frac{\varepsilon}{\text{Lip} f}$$

$$\Rightarrow |f(x) - f(y)| \leq \text{Lip} f \cdot d(x, y) < \varepsilon$$

Thus, by Snapping lemma,

$\exists$ cycle $\bar{z}(\alpha)$ in ℓ -skeleton of K that

$$\Rightarrow f(x) < f(y) + \varepsilon \Rightarrow f(x) < b(\alpha) + \varepsilon \Rightarrow x \in X_{b(\alpha)+\varepsilon}$$

is homologous to $z(\alpha)$ in $X_{b(\alpha)+\varepsilon}$.

$$\text{Also } \alpha \cdot \text{Lip} f = \varepsilon \text{ \& so } x \in X_{b(\alpha)}^{\alpha} \Rightarrow x \in X_{b(\alpha)+\varepsilon}$$

Using this construction, $\bar{z}_i = \bar{z}(\alpha_i)$ for each pt in $D_{\text{pse}, \varepsilon}(f)$ whose persistence exceeds ε , α_i being an ℓ -dim homology class associated to that point.

Now,

let $n \rightarrow$ no. of pts in the ℓ^{th} diagram with pers exceeding ε

Assume indices are chosen s.t. $b(\alpha_1) \leq b(\alpha_2) \leq \dots \leq b(\alpha_n)$

Crucial prop of this construction: Independence of n cycles in the ℓ -skeleton of K

$\hookrightarrow$ Pf by induction:

Spce $\bar{z}_i$ is not ind of its predecessors. Then it is homologous to linear combination

$$\bar{z}_i \sim \sum_{j=1}^{i-1} \lambda_j \bar{z}_j$$

By construction, we have $\bar{z}_j \sim z(\alpha_j)$ in $X_{b(\alpha_j)+\varepsilon}$ for each $1 \leq j \leq i$.

$\therefore b(d_j) \leq b(d_i)$ for each j ,

$\therefore \bar{z}_j \sim \bar{z}(d_j)$ in $X_{b(d_j)+\epsilon}$

Thus,

$$\bar{z}(d_i) \sim \sum_{j=1}^{i-1} \lambda_j \bar{z}(d_j) \text{ in } X_{b(d_i)+\epsilon}$$

Now let $k \leq i$ be the smallest index s.t. $b(d_k) = b(d_{k+1}) = \dots = b(d_i)$. We have a non-zero combination of $\bar{z}(d_k)$ to $\bar{z}(d_i)$ homologous to $\sum_{j=1}^{k-1} \lambda_j \bar{z}(d_j)$. By construction, the death of the corresponding combination of classes d_k to d_i is no earlier than the death of d_i .

This $\times$ the assumption that $d(d_i) > b(d_i) + \epsilon$

Now,

being independent, the no. of cycles $\bar{z}_i$ is at most rank of the l^{th} homology group of the l -skeleton, which is $\leq l$ simplices, $n \leq \text{rank } H_l(K^{(n)}) \leq N_k(n)$

$\therefore$ Inequality follows as $N(n) \geq \sum_k N_k(n)$

Persistence Moments

let $K \in R^+ U \text{ to } \mathbb{R}$

$$\text{Pers}_k(f, t) = \sum_{\text{persistence} > t} \text{pers}(x)^k$$

If $t=0$,

$$\text{Pers}_k(f) = \text{Pers}_k(f, 0) \rightarrow \text{degree-}k \text{ total persistence of } f$$

Moment Lemma: let $X \rightarrow \text{trian}$, cft & $f \rightarrow$ tame Lipschitz from X to $\mathbb{R}$. Then, $\text{Pers}_k(f, t)$

is bdd from above by

$$t^k N\left(\frac{t}{\text{lip}(f)}\right) + k \int_{\epsilon=t}^{\text{amp}(f)} N\left(\frac{\epsilon}{\text{lip}(f)}\right) \epsilon^{k-1} d\epsilon$$

Pf:

let $P(\epsilon)$ be the no. of pts in the diagram of f whose persistence exceeds ϵ .

By the Persistent Cycle lemma,

$$P(\epsilon) \leq N\left(\frac{\epsilon}{\text{lip}(f)}\right)$$

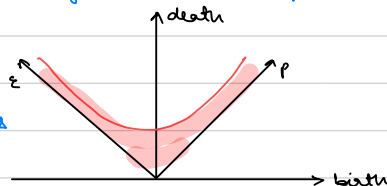
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$$\# \{ (b_i, d_i) \in \text{Dgm}_k(f) : d_i - b_i \geq \epsilon \}$$

Now, as $\varepsilon \uparrow$, fewer features survive past persistence ε & so $P(\varepsilon)$ is non-increasing and decrease by 1 at each persistence value $\varepsilon_i = d_i - b_i$

$\therefore$ using the jump formula, the derivative of P in the sense of distributions is $\frac{dP}{d\varepsilon} = - \sum_i \delta(\varepsilon - \varepsilon_i)$

By projecting diagram to anti-diagonal, every pt (b, d) in the persistence diagram lies above the diagonal. Projecting onto the anti-diagonal line $d=b$ gives us the persistence ε . The Euclidean distance from (b, d) to the diagonal is $\frac{1}{\sqrt{2}}(d-b)$, and so we rescale by $\sqrt{2}$ to get the persistence



Now,

the k^{th} moment of persistences exceeding t is :

$$\text{Per}_k(f, t) = \sum_{\varepsilon_i \geq t} \varepsilon_i^k = \int_t^{\infty} \varepsilon^k \left(-\frac{dP}{d\varepsilon} \right) d\varepsilon$$

By IBP,

$$\text{Per}_k(f, t) = \left[-P(\varepsilon) \varepsilon^k \right]_t^{\infty} + \int_t^{\infty} P(\varepsilon) \cdot k \varepsilon^{k-1} d\varepsilon$$

for $\varepsilon > \text{Amp}(f)$, $P(\varepsilon)$ vanishes and so

$$\text{Per}_k(f, t) \leq t^k P(t) + k \int_t^{\infty} P(\varepsilon) \cdot \varepsilon^{k-1} d\varepsilon$$

Applying Persistent Cycle lemma, we conclude.

Polynomial Growth and Bounded Total Persistence

Assume that the size of the smallest triangulation grows polynomially with one over the mesh i.e assume $\exists C_0$ and $M : N(x) \leq \frac{C_0}{x^M}$ for every $x > 0$

Let A & B be 2 upper bounds in the Moment Lemma i.e $\text{Per}_k(f, t) \leq A + B$

Assuming polynomial growth and setting $k = M + s$ for some $s > 0$, we can find constant upper bounds for both terms :

$$\begin{aligned} A &\leq t^{M+s} C_0 \frac{\text{Lip}(f)^M}{t^M} \leq C_0 \text{Lip}(f)^M \text{Amp}(f)^s \quad \text{and} \\ B &\leq (M+s) \int_t^{\text{Amp}(f)} \frac{C_0 \text{Lip}(f)^M}{\varepsilon^M} \varepsilon^{M+s-1} d\varepsilon = C_0 \text{Lip}(f)^M (M+s) \left[\frac{1}{s} \varepsilon^s \right]_t^{\text{Amp}(f)} \\ &\leq C_0 \text{Lip}(f)^M \text{Amp}(f)^s \frac{M+s}{s} \end{aligned}$$

Both u.b.s depend only on X , $\text{Lip}(f)$ and chosen constant s .

Def: A metric space X implies bounded degree- k total persistence if $\exists C_X$ s.t. $\text{Pers}_k(f) \leq C_X$ for every tame fn $f: X \rightarrow \mathbb{R}$ with $\text{Lip}(f) \leq 1$

Ex: 1) for general Lipschitz fns,

$$\text{Pers}_k(f) \leq C \text{ where } C = C_X \text{Lip}(f)^k$$

2) $X = S^n$

$$\text{Then } N(X) \leq \frac{C_0}{2^n} \Rightarrow \text{Pers}_k(f) \leq C \quad \forall k = n+s \text{ where } s > 0$$

This gives $k = 1+s$ for the circle S^1 .

↳ can be lowered to $k=1$

Results

Wasserstein Distance: $f, g: X \rightarrow \mathbb{R}$ tame fns with $Dgm_k(f)$ & $Dgm_k(g)$

Assuming bounded degree- k total persistence

degree- p Wasserstein distance:

$$W_p(f, g) = \left[\sum_x \inf_{\gamma_x} \sum_{\gamma_x \in Dgm_k(f)} \|x - \gamma_x(x)\|_\infty^p \right]^{1/p}$$

where

$\gamma_x: Dgm_k(f) \rightarrow Dgm_k(g)$ are bijections

Wasserstein Stability Thm: $X \rightarrow$ triangulable, cpt metric space that implies bdd degree- k

total persistence for some real number $k \geq 1$, and let

$f, g: X \rightarrow \mathbb{R}$ be tame Lipschitz fns. Then,

$$W_p(f, g) \leq C^{1/p} \cdot \|f - g\|_\infty^{1-k/p} \quad \forall p \geq k \text{ where}$$

$$C = C_X \max\{\text{Lip}(f)^k, \text{Lip}(g)^k\}, \quad W_0(Dgm_k(f), Dgm_k(g)) \leq \|f - g\|_\infty \leq \varepsilon$$

Pf:

↳ Existence from bottleneck stability thm

let $\gamma_x: Dgm_k(f) \rightarrow Dgm_k(g)$ be the bijection s.t. $\|x - \gamma_x(x)\|_\infty \leq \varepsilon = \|f - g\|_\infty$ for each pt x in the first diagram.

↳ Additionally, we require that $\|x - \gamma_x(x)\|_\infty \leq \frac{1}{2} [\text{pers}(x) + \text{pers}(\gamma_x(x))]$

↳ If not, then $\text{pers}(x) < 2\varepsilon$ & $\text{pers}(\gamma_x(x)) < 2\varepsilon$ and we can change the bijection

by matching both points with pte on the diagonal within L_0 -distance ε .

Thus,

$$\begin{aligned} W_p(f, g)^p &\leq \sum_{i, x} \|x - \chi_i(x)\|_\infty^p \quad (\text{by def}) \\ &\leq \varepsilon^{p \cdot K} \sum_{i, x} \|x - \chi_i(x)\|_\infty^K \quad (\because \|x - \chi_i(x)\|_\infty \leq \varepsilon \ \& \ a \leq \varepsilon \Rightarrow a^p \leq \varepsilon^{p \cdot K} a^K) \\ &\leq \frac{\varepsilon^{p \cdot K}}{2^K} \sum_{i, x} [\text{pos}(x) + \text{pos}(\chi_i(x))]^K \quad (\because \|x - \chi_i(x)\|_\infty \leq \frac{1}{2} (\text{pos}(x) + \text{pos}(\chi_i(x))) \\ &\quad - (i)) \end{aligned}$$

Now,

we know that if ψ is convex, then $\psi\left(\frac{u+v}{2}\right) \leq \frac{1}{2} [\psi(u) + \psi(v)]$

As $a \mapsto a^k$ is convex,

setting $u = 2 \operatorname{per}(x)$ & $v = 2 \operatorname{per}(x_1(x))$,

$$(i) \leq \sum_{j=k+1}^{p-k} \sum_{x \in X} [(2 \text{pens}(x))^k + (2 \text{pens}(Y_k(x)))^k] - C_2$$

New as X implies bnd^Z degree- k total persistence,

$$C_2) \subseteq C \varepsilon^{p-k} \quad \text{where } C = C_{\times} \max \{ \text{Lip}(f)^k, \text{Lip}(g)^k \}$$

Total Persistence Stability Thm: $X \rightarrow$ triangulable, cpt metric space that implies degree- k total persistence for some real number $k \geq 0$, and let $f, g: X \rightarrow \mathbb{R}$ be same Lipschitz for Thm.

$$|P_{\text{sup}}(f) - P_{\text{sup}}(g)| \leq L p w^{p-1-k} C \cdot \|f - g\|_{\infty} \quad \forall p \geq k+1 \quad \text{where}$$

$$C = C_x \max\{Lip(f)^k, Lip(g)^k\} \text{ and } w \leq \max\{Amp(f), Amp(g)\}$$

Pf :

$$y^p - x^p = \int_x^y p z^{p-1} dz \leq p|y-x| \cdot \max\{x,y\}^{p-1} \quad \forall x,y \geq 0 \text{ \& } p \geq 1$$

Using the $\mathbb{Z}$ bottleneck stability thm, we can index the persistences of the points in the diagrams f & g s.t

$$\|p\|_k(f) = p_1^k + \dots + p_m^k \leq C \text{ and}$$

$$\text{Per}_k(q) = q_1^k + \dots + q_m^k \leq C \quad \&$$

$$|p_i - q_i| \leq 2\varepsilon \quad \forall i \text{ where } \varepsilon = \|f - g\|_\infty, \text{ possibly after adding zeroes}$$

Let $w = \max \{p_i, q_i\}$ and writing $\Delta = \text{Per}_p(f) - \text{Per}_p(g)$, we get

$$|A| \leq \sum_{i=1}^m |p_i^p - q_i^p| \leq \sum_{i=1}^m p |p_i - q_i| \max_{i=1, \dots, m} s_{p_i, q_i} z^{p-1} \\ \leq p \cdot 2\epsilon \cdot w^{p-1-k} \underbrace{\sum_{i=1}^m \max_{i=1, \dots, m} s_{p_i, q_i} s^k}_{\sum_{i=1}^m (p_i^k + q_i^k)} \leq 2C$$